Procedures for determining the slope and deflection of a beam

1. Sketch free-body diagrams and develop a **Moment function** $M(x)$ for each interval of the beam.

2. Integrate the second-order differential equation (the moment-curvature equation, Eq. 8.1) for each interval. This introduces some integration constants.

3. Write down the appropriate **boundary conditions** and **continuity conditions**.

4. Use the boundary and continuity conditions of Step 3 to **evaluate the integration constants** introduced in Step 2.

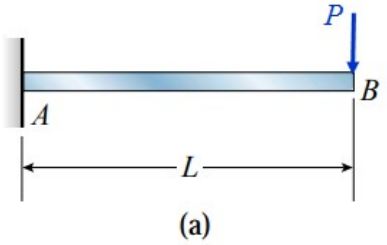
5. Complete the solution by evaluating the slope and deflection at points of interest, by evaluating deflection maxima, etc.

$$\frac{\mathrm{d}^2 v}{\mathrm{d}x^2} = \frac{M}{EI} \qquad (8.1)$$

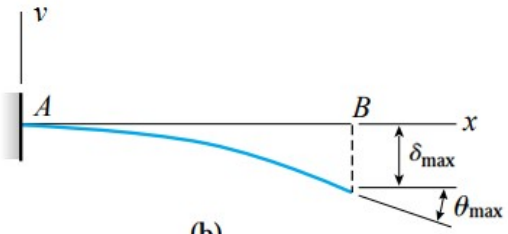
$$\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{EI}\int M\mathrm{d}x + C_1 \qquad (8.2)$$

$$v = \frac{1}{EI}\int\int M\mathrm{d}x\mathrm{d}x + C_1 x + C_2 \qquad (8.3)$$

Example 4. Find the deflection curve and the maximum deflection of the cantilever beam with a concentrated load at the free end shown in Fig. a. What is the slope at the free end?



(a)



(b)

$$\frac{dv}{dx} = \frac{1}{EI}\int M dx + C_1 \qquad (8.2)$$

Solution

Using a free-body diagram (Chapter 8), the internal bending moment of the beam can be obtained as

$$M = P(x - L) \qquad \text{(a)}$$

Substituting the bending moment into Eq. (8.2) and integrating:

$$\frac{dv}{dx} = \frac{P}{2EI}(x^2 - 2Lx) + C_1 \qquad \text{(b)}$$

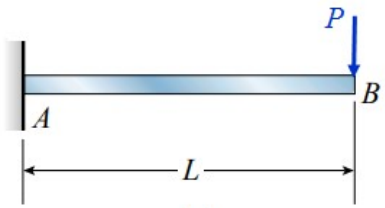
It is the slope function of the beam. The deflection function can be obtained by integrating Eq. (b). To give:

$$v = \frac{P}{6EI}(x^3 - 3Lx^2) + C_1 x + C_2 \qquad \text{(c)}$$

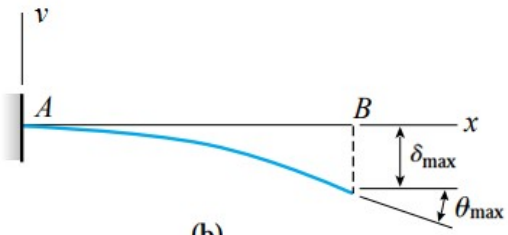
As illustrated in the deflection diagram (Fig. b), the deflection is downward over the entire beam, with the maximum deflection and slope occurring at B. The boundary conditions for this problem are: The deflection and slope at the fixed end are zero:

at $x = 0$: $v = 0$ and $dv/dx = 0$;

So, $C_1 = 0$ and $C_2 = 0$



(a)



(b)

Substitute the constants C_1 and C_2 into Eqs. (b) and (c), we can get the slope and deflection functions of the beam as

$$\frac{dv}{dx} = \frac{P}{2EI}(x^2 - 2Lx) \qquad \text{(d)}$$

$$v = \frac{P}{6EI}(x^3 - 3Lx^2) \qquad \text{(e)}$$

The maximum deflection is at the free end of the beam, $x = L$:

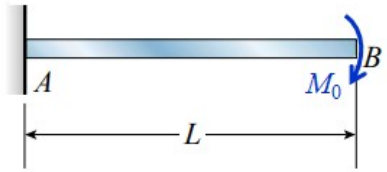
$$\delta_{max} = v\Big|_{x=L} = -\frac{PL^3}{3EI} \qquad \text{(f)}$$

The slope at the free end $x = L$ is

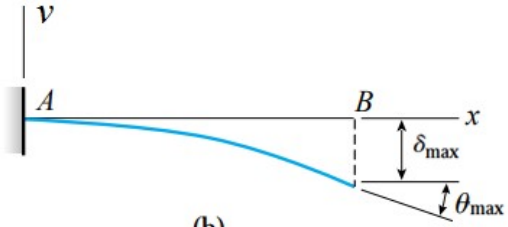
$$\theta_{max} = \frac{dv}{dx}\Big|_{x=L} = -\frac{PL^2}{2EI} \qquad \text{(g)}$$

It is the maximum rotation of the beam.

Example 5. Find the deflection curve and the maximum deflection of the cantilever beam with a couple at the free end shown in Fig. a. What is the slope at the free end?



(a)



(b)

$$\frac{dv}{dx} = \frac{1}{EI}\int M dx + C_1 \qquad (8.2)$$

Solution

The bending moment in the beam is constant and is

$$M = -M_0 \qquad \text{(a)}$$

After two successive integrations of Eq. (8.2), the slope and deflection functions can be obtained as

$$\frac{dv}{dx} = -\frac{M_0}{EI}x + C_1 \qquad \text{(b)}$$

$$v = -\frac{M_0}{2EI}x^2 + C_1 x + C_2 \qquad \text{(c)}$$

The beam deflects as shown in Fig. b. The boundary conditions are such that the deflection and slope at the fixed end are zero:

At $x = 0$: $v = 0$ and $dv/dx = 0$.

So, $C_1 = 0$ and $C_2 = 0$

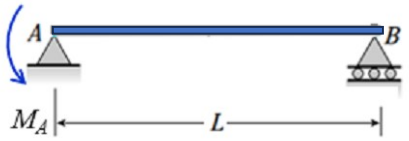
Therefore, the deflection curve of the beam is

$$v = -\frac{M_0}{2EI}x^2 \qquad \text{(c)}$$

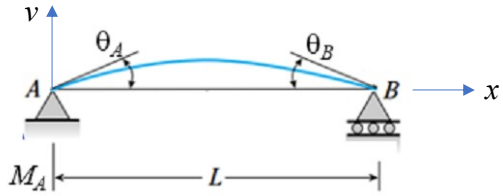
The maximum deflection and slope are at the free end ($x = L$). They are

$$\delta_{max} = v|_{x=L} = -\frac{M_0 L^2}{2EI}, \quad \theta_{max} = \frac{dv}{dx}\Big|_{x=L} = -\frac{M_0 L}{EI}$$

Example 6. Find the deflection curve and the maximum deflection of a simple beam AB with a moment M_A applied at end A (Fig. a). What is the maximum deflection.



(a)



(b)

$$\frac{dv}{dx} = \frac{1}{EI}\int M dx + C_1 \qquad (8.2)$$

Solution

Using a free-body diagram method (Chapter 8), the internal bending moment of the beam can be established as

$$M = \frac{M_A}{L}(x - L) \qquad \text{(a)}$$

Equation (8.2) is integrated with the substitution of Eq. (a). To give the slope function of the beam:

$$\frac{dv}{dx} = \frac{M_A}{2EIL}(x^2 - 2Lx) + C_1 \qquad \text{(b)}$$

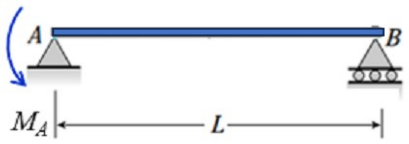
Equation (b) is integrated to give the deflection function of the beam:

$$v = \frac{M_A}{6EIL}(x^3 - 3Lx^2) + C_1 x + C_2 \qquad \text{(c)}$$

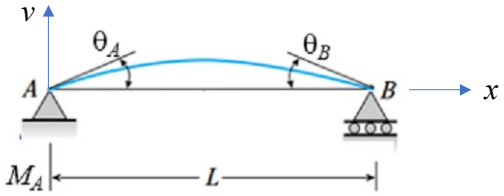
We need two boundary conditions to determine the constants C_1 and C_2. The beam deflects as shown in Fig. b. The boundary conditions require zero displacement at points A and B.

(1) at $x = 0$: $v = 0$; So, $C_2 = 0$

(2) at $x = L$: $v = 0$; So, $C_1 = \frac{M_0 L}{3EI}$



(a)



(b)

Substituting the constants C_1 and C_2 into Eq. (b), we can obtain the deflection of the beam as

$$v = \frac{M_A}{6EIL}(x^3 - 3Lx^2 + 2L^2x) \qquad \text{(d)}$$

Maths unit tells us that the peak value of function (d) is at $\frac{dv}{dx} = 0$.

$$\frac{dv}{dx} = \frac{M_A}{6EIL}(3x^2 - 6Lx + 2L^2)$$

So, $\quad 3x^2 - 6Lx + 2L^2 = 0$

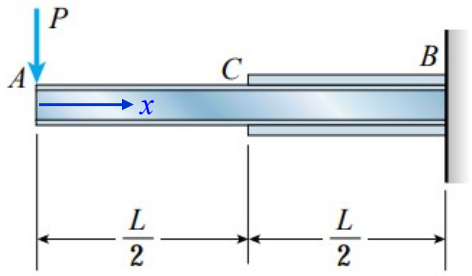
So. $\quad x = 0.4226L$

We see that, for this problem, the maximum deflection is **NOT** at the mid-point of the beam.

The maximum deflection is obtained from Eq. (d), with the substitution of $x = 0.4226\,L$ as:

$$\delta_{\max} = v|_{x=0.4226L} = \frac{0.06415M_AL^2}{EI}$$

Example 7. A cantilever beam ACB has length L and two different values of flexural rigidity, $(EI)_1$, $(EI)_2$. The beam supports a concentrated load P at the free end A. Determine the deflection curve of the beam. What is the deflection at the free end (δ_A)?



$$\frac{dv}{dx} = \frac{1}{EI}\int M dx + C_1 \qquad (8.2)$$

Solution

Since this beam has two intervals, we need two deflection curves, namely, curve for AC and curve for CB, where x is directed to the right from A.

(1) Beam AC (denoted as #1): The bending moment is $M_1 = -Px$

Based on Eq. (8.2), the slopes and deflection for AC can be obtained as

$$\frac{dv_1}{dx} = -\frac{Px^2}{2(EI)_1} + C_1 \qquad \text{(a)}$$

$$v_1 = -\frac{Px^3}{6(EI)_1} + C_1 x + C_2 \qquad \text{(b)}$$

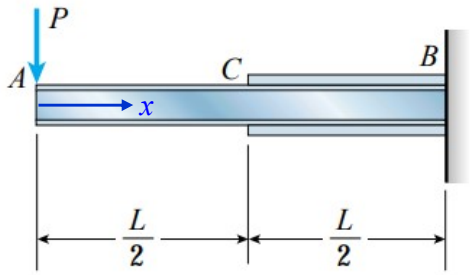
(2) Beam CB (denoted as #2): The bending moment is $M_2 = -Px$

The slope and deflection for segment CB can be obtained as

$$\frac{dv_2}{dx} = -\frac{Px^2}{2(EI)_2} + C_3 \qquad \text{(c)}$$

$$v_2 = -\frac{Px^3}{6(EI)_2} + C_3 x + C_4 \qquad \text{(d)}$$

Four boundary and continuity conditions are needed to determine the four unknown constants (C_1, C_2, C_3, C_4).



The boundary conditions at the fixed end are obvious: the deflection and slope are zero:

at $x = L$: $v = 0$ and $dv_2/dx = 0$;

So, $$-\frac{PL^3}{6(EI)_2} + C_3L + C_4 = 0 \qquad \text{(e)}$$

$$-\frac{PL^2}{2(EI)_2} + C_3 = 0 \qquad \text{(f)}$$

The two regions are joined securely at $x = x/2$. As such, the deflections of the two segments are equal at point C. Same for the slope. Therefore,

at $x = L/2$: $v_1 = v_2$ and $dv_1/dx = dv_2/dx$

Substitution of Eqs. (a) and (b) into above, we get:

$$-\frac{PL^3}{48(EI)_1} + \frac{L}{2}C_1 + C_2 = -\frac{PL^3}{48(EI)_2} + \frac{L}{2}C_3 + C_4 \qquad \text{(g)}$$

$$-\frac{PL^2}{8(EI)_1} + C_1 = -\frac{PL^2}{8(EI)_2} + C_3 \qquad \text{(h)}$$

Solving Eqs. (e) through (h), we obtain

$$C_1 = \frac{3PL^2}{8(EI)_2} + \frac{PL^2}{8(EI)_1},\ C_2 = -\frac{7PL^3}{24(EI)_2} - \frac{PL^3}{24(EI)_1}$$

$$C_3 = \frac{PL^2}{2(EI)_2},\ C_4 = -\frac{PL^3}{3(EI)_2}$$

Having obtained these constants, the deflection curve for the beam can be obtained from Eqs. (b) and (d) as

$$v_1 = -\frac{Px^3}{6(EI)_1} + \left(\frac{3PL^2}{8(EI)_2} + \frac{PL^2}{8(EI)_1}\right)x - \frac{7PL^3}{24(EI)_2} - \frac{PL^3}{24(EI)_1}$$

$$v_2 = -\frac{Px^3}{6(EI)_2} + \frac{PL^2}{2(EI)_2}x - \frac{PL^3}{3(EI)_2}$$

In particular, the deflection at the free end is

$$\delta_0 = v_1\Big|_{x=0} = -\frac{7PL^3}{24(EI)_2} - \frac{PL^3}{24(EI)_1}$$

If the two segments of the beam are same, $(EI)_1 = (EI)_2 = EI$. The deflection is

$$\delta_0 = v_1\Big|_{x=0} = -\frac{PL^3}{3EI}$$

We have got this result in example 4.

Appendix:

Below differentiations are frequently used:

- if $v = A$, $\dfrac{\mathrm{d}v}{\mathrm{d}x} = 0$
- if $v = Ax$, $\dfrac{\mathrm{d}v}{\mathrm{d}x} = A$
- if $v = Ax^2$, $\dfrac{\mathrm{d}v}{\mathrm{d}x} = 2Ax$
- if $v = Ax^3$, $\dfrac{\mathrm{d}v}{\mathrm{d}x} = 3Ax^2$
- if $v = Ax^4$, $\dfrac{\mathrm{d}v}{\mathrm{d}x} = 4Ax^3$

A is constant

Below integrals are frequently used:

- If $M = M_0$, $\int M\mathrm{d}x = M_0 x + A$,
- If $M = Px$, $\int M\mathrm{d}x = \dfrac{P}{2}x^2 + A$,
- If $M = qx^2$, $\int M\mathrm{d}x = \dfrac{q}{3}x^3 + A$,
- If $M = \omega x^3$, $\int M\mathrm{d}x = \dfrac{\omega}{4}x^4 + A$,

A is constant

You can also look up tables – (e.g., Tables of integrals, series and products)

Quick review : Deflection of beam

Integration of the second-order differential equation – *second-order method*

$$\frac{\mathrm{d}^2 v}{\mathrm{d}x^2} = \frac{M}{EI} \qquad (8.1)$$

If, by any means, the deflection has been obtained, the internal bending moment can be obtained from Eq. (8.1) as

$$M = EI\frac{\mathrm{d}^2 v}{\mathrm{d}x^2} \qquad (8.4)$$

And the transverse shear force can be obtained from Eq. (8.2) of Chapter 8:

$$V = \frac{dM}{\mathrm{d}x} \qquad (8.2)$$